

# Integration Manual

for S32K1\_S32M24X LIN\_43\_LPUART\_FLEXIO Driver

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## Chapter 1

### Revision History

| Revision | Date       | Author       | Description                                                  |
|----------|------------|--------------|--------------------------------------------------------------|
| 1.0      | 07.03.2025 | NXP RTD Team | S32K1_S32M24X Real-Time Drivers AUTOSAR R21-11 Version 3.0.0 |

## Chapter 2

### Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This integration manual describes the integration requirements for LIN Driver for S32K1\_S32M24X microcontrollers.

### 2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k116\_qfn32
- s32k116\_lqfp48
- s32k118\_lqfp48
- s32k118\_lqfp64
- s32k142\_lqfp48
- s32k142\_lqfp64
- s32k142\_lqfp100
- s32k142w\_lqfp48
- s32k142w\_lqfp64
- s32k144\_lqfp48
- s32k144\_lqfp64 / MWCT1014S\_lqfp64
- s32k144\_lqfp100 / MWCT1014S\_lqfp100
- s32k144\_mapbga100
- s32k144w\_lqfp48
- s32k144w\_lqfp64
- s32k146\_lqfp64
- s32k146\_lqfp100 / MWCT1015S\_lqfp100

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- s32k146\_mapbga100 / MWCT1015S\_mapbga100
- s32k146\_lqfp144
- s32k148\_lqfp100
- s32k148\_mapbga100 / MWCT1016S\_mapbga100
- s32k148\_lqfp144
- s32k148\_lqfp176
- s32m241\_lqfp64
- s32m242\_lqfp64
- s32m243\_lqfp64
- s32m244\_lqfp64

All of the above microcontroller devices are collectively named as S32K1\_S32M24X. Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power

## 2.2 Overview

**AUTOSAR (AUTomotive Open System ARchitecture)** is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

## 2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

## 2.4 Acronyms and Definitions

| Term  | Definition                               |
|-------|------------------------------------------|
| API   | Application Programming Interface        |
| ASM   | Assembler                                |
| BSMI  | Basic Software Make file Interface       |
| CAN   | Controller Area Network                  |
| C/CPP | C and C++ Source Code                    |
| CS    | Chip Select                              |
| CTU   | Cross Trigger Unit                       |
| DEM   | Diagnostic Event Manager                 |
| DET   | Development Error Tracer                 |
| DMA   | Direct Memory Access                     |
| ECU   | Electronic Control Unit                  |
| FIFO  | First In First Out                       |
| LSB   | Least Significant Bit                    |
| MCU   | Micro Controller Unit                    |
| MIDE  | Multi Integrated Development Environment |
| MSB   | Most Significant Bit                     |
| N/A   | Not Applicable                           |
| RAM   | Random Access Memory                     |
| SIU   | Systems Integration Unit                 |
| SWS   | Software Specification                   |
| VLE   | Variable Length Encoding                 |
| XML   | Extensible Markup Language               |

## 2.5 Reference List

| #  | Title                    | Version                                                 |
|----|--------------------------|---------------------------------------------------------|
| 1  | Specification of Driver  | AUTOSAR Release R21-11                                  |
| 2  | S32K1xx Reference Manual | Rev.14.1, 01/2024                                       |
| 3  | S32M24x Reference Manual | Rev. 5, 12/2024                                         |
| 4  | S32K1xx Data Sheet       | Rev. 14, 08/2021                                        |
| 5  | S32M2xx Data Sheet       | Rev. 7, 12/2024                                         |
| 6  | S32K116 Errata           | Mask Set Errata for Mask 0N96V Rev. 17/JAN/2024, 1/2024 |
| 7  | S32K118 Errata           | Mask Set Errata for Mask 0N97V Rev. 15/JAN/2024, 1/2024 |
| 8  | S32K142 Errata           | Mask Set Errata for Mask 0N33V Rev. 15/JAN/2024, 1/2024 |
| 9  | S32K144 Errata           | Mask Set Errata for Mask 0N57U Rev. 15/JAN/2024, 1/2024 |
| 10 | S32K144W Errata          | Mask Set Errata for Mask 0P64A Rev. 03/JUL/2024, 7/2024 |
| 11 | S32K146 Errata           | Mask Set Errata for Mask 0N73V Rev. 15/JAN/2024, 1/2024 |
| 12 | S32K148 Errata           | Mask Set Errata for Mask 0N20V Rev. 15/JAN/2024, 1/2024 |
| 13 | S32M242 Errata           | Mask Set Errata for Mask N33V+P69K Rev.1, 8/2024        |
| 14 | S32M244 Errata           | Mask Set Errata for Mask P64A+P69K Rev.1, 8/2024        |

## Chapter 3

### Building the driver

- [Build Options](#)
- [Files required for compilation](#)
- [Setting up the plugins](#)

This section describes the source files and various compilers, linker options used for building the driver. It also explains the EB Tresos Studio plugin setup procedure.

#### 3.1 Build Options

- [GCC Compiler/Assembler/Linker Options](#)
- [IAR Compiler/Assembler/Linker Options](#)
- [GHS Compiler/Assembler/Linker Options](#)

The RTD driver files are compiled using:

- NXP GCC 10.2.0 20200723 (Build 1728 Revision g5963bc8)
- IAR ANSI C/C++ Compiler V8.40.3.228/W32 for ARM Functional Safety
- Green Hills Multi 7.1.6d / Compiler 2020.1.4

The compiler, assembler, and linker flags used for building the driver are explained below.

The TS\_T40D2M30I0R0 part of the plugin name is composed as follows:

- T = Target\_Id (e.g. T40 identifies Cortex-M architecture)
- D = Derivative\_Id (e.g. D2 identifies S32K1 platform)
- M = SW\_Version\_Major and SW\_Version\_Minor
- I = SW\_Version\_Patch
- R = Reserved

##### 3.1.1 GCC Compiler/Assembler/Linker Options

###### 3.1.1.1 GCC Compiler Options

| Compiler Option | Description                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| -mcpu=cortex-m4 | Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices) |



| Compiler Option                       | Description                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -mcpu=cortex-m0plus                   | Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)                                                                                                                                 |
| -mthumb                               | Generates code that executes in Thumb state                                                                                                                                                                                        |
| -mlittle-endian                       | Generate code for a processor running in little-endian mode                                                                                                                                                                        |
| -mfpv4-sp-d16                         | Specifies the floating-point hardware available on the target (for S32K14x or S32M24x devices)                                                                                                                                     |
| -mfloat-abi=hard                      | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x or S32M24x devices)                                                        |
| -mfpv4-sp-d16                         | Specifies the floating-point hardware available on the target (for S32K11x devices)                                                                                                                                                |
| -mfloat-abi=soft                      | Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices)                                                              |
| -std=c99                              | Specifies the ISO C99 base standard                                                                                                                                                                                                |
| -Os                                   | Optimize for size. Enables all -O2 optimizations except those that often increase code size                                                                                                                                        |
| -ggdb3                                | Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program |
| -Wall                                 | Enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros                                                 |
| -Wextra                               | This enables some extra warning flags that are not enabled by -Wall                                                                                                                                                                |
| -pedantic                             | Issue all the warnings demanded by strict ISO C. Reject all programs that use forbidden extensions. Follows the version of the ISO C standard specified by the aforementioned -std option                                          |
| -Wstrict-prototypes                   | Warn if a function is declared or defined without specifying the argument types                                                                                                                                                    |
| -Wundef                               | Warn if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero                                                                                                                          |
| -Wunused                              | Warn whenever a function, variable, label, value, macro is unused                                                                                                                                                                  |
| -Werror=implicit-function-declaration | Make the specified warning into an error. This option throws an error when a function is used before being declared                                                                                                                |
| -Wsign-compare                        | Warn when a comparison between signed and unsigned values could produce an incorrect result when the signed value is converted to unsigned.                                                                                        |
| -Wdouble-promotion                    | Give a warning when a value of type float is implicitly promoted to double                                                                                                                                                         |
| -fno-short-enums                      | Specifies that the size of an enumeration type is at least 32 bits regardless of the size of the enumerator values.                                                                                                                |
| -funsigned-char                       | Let the type char be unsigned by default, when the declaration does not use either signed or unsigned                                                                                                                              |
| -funsigned-bitfields                  | Let a bit-field be unsigned by default, when the declaration does not use either signed or unsigned                                                                                                                                |

| Compiler Option                 | Description                                                                                                                                                                                                                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -fomit-frame-pointer            | Omit the frame pointer in functions that don't need one. This avoids the instructions to save, set up and restore the frame pointer; on many targets it also makes an extra register available.                                                                                       |
| -fno-common                     | Makes the compiler place uninitialized global variables in the BSS section of the object file. This inhibits the merging of tentative definitions by the linker so you get a multiple-definition error if the same variable is accidentally defined in more than one compilation unit |
| -fstack-usage                   | This option is only used to build test for generation Ram/↔ Stack size report. Makes the compiler output stack usage information for the program, on a per-function basis                                                                                                             |
| -fdump-ipa-all                  | This option is only used to build test for generation Ram/↔ Stack size report. Enables all inter-procedural analysis dumps                                                                                                                                                            |
| -c                              | Stop after assembly and produce an object file for each source file                                                                                                                                                                                                                   |
| -DUSE_SW_VECTOR_MODE            | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode                                                                                                                                         |
| -DI_CACHE_ENABLE                | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32↔M24x devices)                                                                                                  |
| -DENABLE_FPU                    | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)                                                                                                                     |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPORT↔ as a macro, with definition 1. Allows drivers to be configured in user mode.                                                                                                                                                                 |
| -sysroot=                       | Specifies the path to the sysroot, for Cortex-M7 it is /arm-none-eabi/newlib                                                                                                                                                                                                          |
| -specs=nano.specs               | Use Newlib nano specs                                                                                                                                                                                                                                                                 |
| -specs=nosys.specs              | Do not use printf/scanf                                                                                                                                                                                                                                                               |

## 3.1.1.2 GCC Assembler Options

| Assembler Option     | Description                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -Xassembler-with-cpp | Specifies the language for the following input files (rather than letting the compiler choose a default based on the file name suffix)                           |
| -mcpu=cortex-m4      | Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices)                                                    |
| -mcpu=cortex-m0plus  | Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)                                                               |
| -mfpu=fpv4-sp-d16    | Specifies the floating-point hardware available on the target (for S32K14x devices)                                                                              |
| -mfloat-abi=hard     | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x devices) |
| -mfpu=auto           | Specifies the floating-point hardware available on the target (for S32K11x devices)                                                                              |

| Assembler Option | Description                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -mfloat-abi=soft | Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices) |
| -mthumb          | Generates code that executes in Thumb state                                                                                                                           |
| -c               | Stop after assembly and produce an object file for each source file                                                                                                   |

### 3.1.1.3 GCC Linker Options

| Linker Option        | Description                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -Wl,-Map,filename    | Produces a map file                                                                                                                                                                                                                |
| -T linkerfile        | Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)                                                                                                                     |
| -entry=Reset_Handler | Specifies that the program entry point is Reset_Handler                                                                                                                                                                            |
| -nostartfiles        | Do not use the standard system startup files when linking                                                                                                                                                                          |
| -mcpu=cortex-m4      | Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices)                                                                                                                      |
| -mcpu=cortex-m0plus  | Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)                                                                                                                                 |
| -mthumb              | Generates code that executes in Thumb state                                                                                                                                                                                        |
| -mfpu=fpv4-sp-d16    | Specifies the floating-point hardware available on the target (for S32K14x or S32M24x devices)                                                                                                                                     |
| -mfloat-abi=hard     | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x or S32M24x devices)                                                        |
| -mfpu=auto           | Specifies the floating-point hardware available on the target (for S32K11x devices)                                                                                                                                                |
| -mfloat-abi=soft     | Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices)                                                              |
| -mlittle-endian      | Generate code for a processor running in little-endian mode                                                                                                                                                                        |
| -ggdb3               | Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program |
| -lc                  | Link with the C library                                                                                                                                                                                                            |
| -lm                  | Link with the Math library                                                                                                                                                                                                         |
| -lgcc                | Link with the GCC library                                                                                                                                                                                                          |
| -n                   | Turn off page alignment of sections, and disable linking against shared libraries                                                                                                                                                  |
| -sysroot=            | Specifies the path to the sysroot, for Cortex-M7 it is /arm-none-eabi/newlib                                                                                                                                                       |
| -specs=nano.specs    | Use Newlib nano specs                                                                                                                                                                                                              |
| -specs=nosys.specs   | Do not use printf/scanf                                                                                                                                                                                                            |

### 3.1.2 IAR Compiler/Assembler/Linker Options

#### 3.1.2.1 IAR Compiler Options

| Compiler Option | Description                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| -cpu=Cortex-M4  | Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices) |

| Compiler Option                 | Description                                                                                                                                                                                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -cpu=Cortex-M0+                 | Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)                                                                                                               |
| -cpu_mode=thumb                 | Generates code that executes in Thumb state                                                                                                                                                                      |
| -endian=little                  | Generate code for a processor running in little-endian mode                                                                                                                                                      |
| -fpu=FPv4-SP                    | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x or S32M24x devices)                                           |
| -fpu=none                       | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)                                                                        |
| -e                              | Enables all IAR C language extensions                                                                                                                                                                            |
| -Ohz                            | Optimize for size. The compiler will emit AEABI attributes indicating the requested optimization goal. This information can be used by the linker to select smaller or faster variants of DLIB library functions |
| -debug                          | Makes the compiler include debugging information in the object modules. Including debug information will make the object files larger                                                                            |
| -no_clustering                  | Disables static clustering optimizations. Static and global variables defined within the same module will not be arranged so that variables that are accessed in the same function are close to each other       |
| -no_mem_idioms                  | Makes the compiler not optimize certain memory access patterns                                                                                                                                                   |
| -no_explicit_zero_opt           | Do not treat explicit initializations to zero of static variables as zero initializations                                                                                                                        |
| -require_prototypes             | Force the compiler to verify that all functions have proper prototypes. Generates an error otherwise                                                                                                             |
| -no_wrap_diagnostics            | Does not wrap long lines in diagnostic messages                                                                                                                                                                  |
| -diag_suppress=Pa050            | Suppresses diagnostic message Pa050                                                                                                                                                                              |
| -DIAR                           | Predefine IAR as a macro, with definition 1                                                                                                                                                                      |
| -DUSE_SW_VECTOR_MODE            | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode.                                                                   |
| -DI_CACHE_ENABLE                | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)                              |
| -DENABLE_FPU                    | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)                                                |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode.                                                                                             |

### 3.1.2.2 IAR Assembler Options

| Assembler Option | Description                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -cpu=Cortex-M4   | Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices)                                               |
| -cpu=Cortex-M0+  | Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)                                                          |
| -fpu=FPv4-SP     | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x devices) |
| -fpu=none        | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)                   |
| -cpu_mode thumb  | Selects the thumb mode for the assembler directive CODE                                                                                                     |
| -g               | Disables the automatic search for system include files                                                                                                      |
| -r               | Generates debug information                                                                                                                                 |

### 3.1.2.3 IAR Linker Options

| Linker Option                | Description                                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -map filename                | Produces a map file                                                                                                                                                    |
| -config linkerfile           | Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)                                                         |
| -cpu=Cortex-M4               | Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices)                                                          |
| -cpu=Cortex-M0+              | Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)                                                                     |
| -fpu=FPv4-SP                 | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x or S32M24x devices) |
| -fpu=none                    | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)                              |
| -entry __start               | Treats __start as a root symbol and start label                                                                                                                        |
| -enable_stack_usage          | Enables stack usage analysis. If a linker map file is produced, a stack usage chapter is included in the map file                                                      |
| -skip_dynamic_initialization | Dynamic initialization (typically initialization of C++ objects with static storage duration) will not be performed automatically during application startup           |
| -no_wrap_diagnostics         | Does not wrap long lines in diagnostic messages                                                                                                                        |

## 3.1.3 GHS Compiler/Assembler/Linker Options

### 3.1.3.1 GHS Compiler Options

| Compiler Option   | Description                                                              |
|-------------------|--------------------------------------------------------------------------|
| -cpu=cortexm4     | Selects target processor: Arm Cortex M4 (for S32K14x or S32M24x devices) |
| -cpu=cortexm0plus | Selects target processor: Arm Cortex M0+ (for S32K11x devices)           |
| -thumb            | Selects generating code that executes in Thumb state                     |

| Compiler Option            | Description                                                                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -fpu=vfpv4_d16             | Specifies hardware floating-point using the v4 version of the VFP instruction set, with 16 double-precision floating-point registers (for S32K14x or S32M24x devices)                                                          |
| -fsingle                   | Use hardware single-precision, software double-precision FP instructions (for S32K14x or S32M24x devices)                                                                                                                      |
| -fsoft                     | Specifies software floating-point (SFP) mode. This setting causes your target to use integer registers to hold floating-point data and use library subroutine calls to emulate floating-point operations (for S32K11x devices) |
| -C99                       | Use (strict ISO) C99 standard (without extensions)                                                                                                                                                                             |
| -ghstd=last                | Use the most recent version of Green Hills Standard mode (which enables warnings and errors that enforce a stricter coding standard than regular C and C++)                                                                    |
| -Osize                     | Optimize for size                                                                                                                                                                                                              |
| -gnu_asm                   | Enables GNU extended asm syntax support                                                                                                                                                                                        |
| -dual_debug                | Generate DWARF 2.0 debug information                                                                                                                                                                                           |
| -G                         | Generate debug information                                                                                                                                                                                                     |
| -keeptempfiles             | Prevents the deletion of temporary files after they are used. If an assembly language file is created by the compiler, this option will place it in the current directory instead of the temporary directory                   |
| -Wimplicit-int             | Produce warnings if functions are assumed to return int                                                                                                                                                                        |
| -Wshadow                   | Produce warnings if variables are shadowed                                                                                                                                                                                     |
| -Wtrigraphs                | Produce warnings if trigraphs are detected                                                                                                                                                                                     |
| -Wundef                    | Produce a warning if undefined identifiers are used in #if preprocessor statements                                                                                                                                             |
| -unsigned_chars            | Let the type char be unsigned, like unsigned char                                                                                                                                                                              |
| -unsigned_fields           | Bitfields declared with an integer type are unsigned                                                                                                                                                                           |
| -no_commons                | Allocates uninitialized global variables to a section and initializes them to zero at program startup                                                                                                                          |
| -no_exceptions             | Disables C++ support for exception handling                                                                                                                                                                                    |
| -no_slash_comment          | C++ style // comments are not accepted and generate errors                                                                                                                                                                     |
| -prototype_errors          | Controls the treatment of functions referenced or called when no prototype has been provided                                                                                                                                   |
| -incorrect_pragma_warnings | Controls the treatment of valid #pragma directives that use the wrong syntax                                                                                                                                                   |
| -c                         | Stop after assembly and produce an object file for each source file                                                                                                                                                            |
| -DUSE_SW_VECTOR_MODE       | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode                                                                                  |
| -DI_CACHE_ENABLE           | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)                                            |

| Compiler Option                 | Description                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -DENABLE_FPU                    | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices) |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode                                               |

### 3.1.3.2 GHS Assembler Options

| Assembler Option           | Description                                                                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -cpu=cortexm4              | Selects target processor: Arm Cortex M4 (for S32K14x or S32M24x devices)                                                                                                                                                       |
| -cpu=cortexm0plus          | Selects target processor: Arm Cortex M0+ (for S32K11x devices)                                                                                                                                                                 |
| -fpu=vfpv4_d16             | Specifies hardware floating-point using the v4 version of the VFP instruction set, with 16 double-precision floating-point registers (for S32K14x devices)                                                                     |
| -fsingle                   | Use hardware single-precision, software double-precision FP instructions (for S32K14x devices)                                                                                                                                 |
| -fsoft                     | Specifies software floating-point (SFP) mode. This setting causes your target to use integer registers to hold floating-point data and use library subroutine calls to emulate floating-point operations (for S32K11x devices) |
| -preprocess_assembly_files | Controls whether assembly files with standard extensions such as .s and .asm are preprocessed                                                                                                                                  |
| -list                      | Creates a listing by using the name and directory of the object file with the .lst extension                                                                                                                                   |
| -c                         | Stop after assembly and produce an object file for each source file                                                                                                                                                            |

### 3.1.3.3 GHS Linker Options

| Linker Option            | Description                                                                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -e Reset_Handler         | Make the symbol Reset_Handler be treated as a root symbol and the start label of the application                                                                                                  |
| -T linker_script_file.ld | Use linker_script_file.ld as the linker script. This script replaces the default linker script (rather than adding to it)                                                                         |
| -map                     | Produce a map file                                                                                                                                                                                |
| -keepmap                 | Controls the retention of the map file in the event of a link error                                                                                                                               |
| -Mn                      | Generates a listing of symbols sorted alphabetically/numerically by address                                                                                                                       |
| -delete                  | Instructs the linker to remove functions that are not referenced in the final executable. The linker iterates to find functions that do not have relocations pointing to them and eliminates them |
| -ignore_debug_references | Ignores relocations from DWARF debug sections when using -delete. DWARF debug information will contain references to deleted functions that may break some third-party debuggers                  |
| -Llibrary_path           | Points to library_path (the libraries location) for thumb2 to be used for linking                                                                                                                 |
| -larch                   | Link architecture specific library                                                                                                                                                                |
| -lstartup                | Link run-time environment startup routines. The source code for the modules in this library is provided in the src/libstartup directory                                                           |

| Linker Option           | Description                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -lind_sd                | Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library (for S32K14x or S32M24x devices) |
| -lind_sf                | Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library (for S32K11x devices)            |
| -v                      | Prints verbose information about the activities of the linker, including the libraries it searches to resolve undefined symbols                                                                                                                     |
| -keep=C40_Ip_AccessCode | Avoid linker remove function C40_Ip_AccessCode from Fls module because it is not referenced explicitly                                                                                                                                              |
| -nostartfiles           | Controls the start files to be linked into the executable                                                                                                                                                                                           |

## 3.2 Files required for compilation

This section describes the include files required to compile, assemble (if assembler code) and link the LIN driver for S32K1\_S32M24X microcontrollers. To avoid integration of incompatible files, all the include files from other modules shall have the same AR\_MAJOR\_VERSION and AR\_MINOR\_VERSION, i.e. only files with the same AUTOSAR major and minor versions can be compiled.

### 3.2.1 LIN Driver Files

- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\src\Lin\_43\_LPUART\_FLEXIO.c
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\src\Lin\_43\_LPUART\_FLEXIO\_ASRExt.c
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\src\Lin\_43\_LPUART\_FLEXIO\_Ipw.c
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\src\Lin\_43\_LPUART\_FLEXIO\_Ip\_Common.c
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\src\Lpuart\_Lin\_Ip\_Hw\_Access.c
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\src\Lpuart\_Lin\_Ip\_Irq.c
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\src\Lpuart\_Lin\_Ip.c
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\src\Flexio\_Lin\_Ip.c
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lin\_43\_LPUART\_FLEXIO.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lin\_43\_LPUART\_FLEXIO\_ASRExt.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lin\_43\_LPUART\_FLEXIO\_Types.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lin\_43\_LPUART\_FLEXIO\_Ipw.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lin\_43\_LPUART\_FLEXIO\_Ipw\_Types.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lin\_43\_LPUART\_FLEXIO\_Ip\_Common.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lpuart\_Lin\_Ip\_Hw\_Access.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lpuart\_Lin\_Ip.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Lpuart\_Lin\_Ip\_Types.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Flexio\_Lin\_Ip\_Irq.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Flexio\_Lin\_Ip\_Types.h
- Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\include\Flexio\_Lin\_Ip.h



### 3.2.2 LIN Driver Generated Files (must be generated by the user using a configuration tool):

- Lin\_43\_LPUART\_FLEXIO\_Cfg.c
- Lin\_43\_LPUART\_FLEXIO\_Cfg.h
- Lin\_43\_LPUART\_FLEXIO\_Defines.h
- Lin\_43\_LPUART\_FLEXIO\_Ipw\_Cfg.h
- Lpuart\_Lin\_Ip\_Cfg.h
- Lpuart\_Lin\_Ip\_Defines.h
- Flexio\_Lin\_Ip\_CfgDefines.h
- Flexio\_Lin\_Ip\_Defines.h
- Flexio\_Lin\_Ip\_Cfg.h
- Lin\_43\_LPUART\_FLEXIO\_[VariantName]\_PBcfg.c
- Lin\_43\_LPUART\_FLEXIO\_Ipw\_[VariantName]\_PBcfg.c
- Lpuart\_Lin\_Ip\_[VariantName]\_PBcfg.c
- Flexio\_Lin\_Ip\_[VariantName]\_PBcfg.c
- Lin\_43\_LPUART\_FLEXIO\_[VariantName]\_PBcfg.h
- Lin\_43\_LPUART\_FLEXIO\_Ipw\_[VariantName]\_PBcfg.h
- Lpuart\_Lin\_Ip\_[VariantName]\_PBcfg.h
- Flexio\_Lin\_Ip\_[VariantName]\_PBcfg.h

### 3.2.3 BASE Files:

- BaseNXP\_TS\_T40D2M30I0R0\include\Lin\_43\_LPUART\_FLEXIO\_GeneralTypes.h
- BaseNXP\_TS\_T40D2M30I0R0\include\Mcal.h
- BaseNXP\_TS\_T40D2M30I0R0\include\Lin\_43\_LPUART\_FLEXIO\_MemMap.h
- BaseNXP\_TS\_T40D2M30I0R0\include\Platform\_Types.h
- BaseNXP\_TS\_T40D2M30I0R0\include\Soc\_Ips.h
- BaseNXP\_TS\_T40D2M30I0R0\include\StandardTypes.h
- BaseNXP\_TS\_T40D2M30I0R0\include\OsIf.h
- BaseNXP\_TS\_T40D2M30I0R0\include\Devassert.h
- BaseNXP\_TS\_T40D2M30I0R0\generate\_PC\include\modules.h
- BaseNXP\_TS\_T40D2M30I0R0\header\S32K1xx\_LPUART.h
- BaseNXP\_TS\_T40D2M30I0R0\header\S32K1xx\_FLEXIO.h
- BaseNXP\_TS\_T40D2M30I0R0\header\S32M24x\_LPUART.h
- BaseNXP\_TS\_T40D2M30I0R0\header\S32M24x\_FLEXIO.h

### 3.2.4 DEM Files:

- Dem\_TS\_T40D2M30I0R0\include\Dem.h
- Dem\_TS\_T40D2M30I0R0\include\Dem\_Types.h
- Dem\_TS\_T40D2M30I0R0\generate\_PC\include\Dem\_IntErrId.h
- Dem\_TS\_T40D2M30I0R0\src\Dem.c

### 3.2.5 DET Files:

- Det\_TS\_T40D2M30I0R0\include\Det.h
- Det\_TS\_T40D2M30I0R0\src\Det.c

### 3.2.6 RTE Files:

- Rte\_TS\_T40D2M30I0R0\include\SchM\_Lin\_43\_LPUART\_FLEXIO.h
- Rte\_TS\_T40D2M30I0R0\src\SchM\_Lin\_43\_LPUART\_FLEXIO.c

### 3.2.7 LINIF Files:

- LinIf\_TS\_T40D2M30I0R0\include\LinIf.h
- LinIf\_TS\_T40D2M30I0R0\src\LinIf.c

### 3.2.8 ECUM folder:

- EcuM\_TS\_T40D2M30I0R0\generate\_PC\include\EcuM\_Cfg.h
- EcuM\_TS\_T40D2M30I0R0\include\EcuM.h
- EcuM\_TS\_T40D2M30I0R0\src\EcuM.c

### 3.2.9 MCL folder:

- Mcl\_TS\_T40D2M30I0R0\generate\_PC\include\Flexio\_Mcl\_Ip\_Cfg.h
- Mcl\_TS\_T40D2M30I0R0\generate\_PC\include\Flexio\_Mcl\_Ip\_CfgDefines.h
- Mcl\_TS\_T40D2M30I0R0\generate\_PC\include\Flexio\_Mcl\_Ip\_Definitions.h
- Mcl\_TS\_T40D2M30I0R0\generate\_PB\include\Flexio\_Mcl\_Ip\_PBcfg.h
- Mcl\_TS\_T40D2M30I0R0\generate\_PB\include\Flexio\_Mcl\_Ip\_PBcfg.c
- Mcl\_TS\_T40D2M30I0R0\include\Mcl.h
- Mcl\_TS\_T40D2M30I0R0\include\Flexio\_Mcl\_Ip\_HwAccess.h
- Mcl\_TS\_T40D2M30I0R0\include\Flexio\_Mcl\_Ip\_Types.h
- Mcl\_TS\_T40D2M30I0R0\include\Flexio\_Mcl\_Ip.h
- Mcl\_TS\_T40D2M30I0R0\src\Mcl.c
- Mcl\_TS\_T40D2M30I0R0\src\Flexio\_Mcl\_Ip\_HwAccess.c
- Mcl\_TS\_T40D2M30I0R0\src\Flexio\_Mcl\_Ip\_Types.c
- Mcl\_TS\_T40D2M30I0R0\src\Flexio\_Mcl\_Ip.c
- Mcl\_TS\_T40D2M30I0R0\src\Flexio\_Mcl\_Ip\_Irq.c

### 3.3 Setting up the plugins

#### 3.3.1 Location of various files inside the Lin module folder:

- VSMD (Vendor Specific Module Definition) file in EB Tresos Studio XDM format:
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\config\Lin\_43\_LPUART\_FLEXIO.xdm
- VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\autosar\Lin\_43\_LPUART\_FLEXIO\_<subderivative←\_name>.epd
- Code Generation Templates for variant aware parameters:
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PB\src\Lin\_43\_LPUART\_FLEXIO\_←Ipw\_PBcfg.c
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PB\src\Lin\_43\_LPUART\_FLEXIO\_←PBcfg.c
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PB\src\Lpuart\_Lin\_Ip\_PBcfg.c
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PB\src\Flexio\_Lin\_Ip\_PBcfg.c
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PB\include\Lin\_43\_LPUART\_FLEX←IO\_Ipw\_PBcfg.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PB\include\Lin\_43\_LPUART\_FLEX←IO\_PBcfg.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PB\include\Lpuart\_Lin\_Ip\_PBcfg.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PB\include\Flexio\_Lin\_Ip\_PBcfg.h
- Code Generation Templates for parameters without variation points:
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\include\Lin\_43\_LPUART\_FLEX←IO\_Cfg.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\include\Lin\_43\_LPUART\_FLEX←IO\_Defines.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\include\Lin\_43\_LPUART\_FLEX←IO\_Ipw\_Cfg.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\include\Lpuart\_Lin\_Ip\_Cfg.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\include\Lpuart\_Lin\_Ip\_Defines.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\include\Flexio\_Lin\_Ip\_CfgDefines.←h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\include\Flexio\_Lin\_Ip\_Defines.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\include\Flexio\_Lin\_Ip\_Cfg.h
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0\generate\_PC\src\Lin\_43\_LPUART\_FLEXIO\_←Cfg.h

#### 3.3.2 Steps to generate the configuration:

1. Copy the following module folders into the Tresos plugins folder:
  - Lin\_43\_LPUART\_FLEXIO\_TS\_T40D2M30I0R0

## Building the driver

- BaseNXP\_TS\_T40D2M30I0R0
  - Dem\_TS\_T40D2M30I0R0
  - Det\_TS\_T40D2M30I0R0
  - EcuC\_TS\_T40D2M30I0R0
  - EcuM\_TS\_T40D2M30I0R0
  - Rte\_TS\_T40D2M30I0R0
  - Resource\_TS\_T40D2M30I0R0
  - Mcu\_TS\_T40D2M30I0R0
  - LinIf\_TS\_T40D2M30I0R0
  - Mcl\_TS\_T40D2M30I0R0
2. Set the desired Tresos Output location folder for the generated sources and header files.
  3. Use the EB Tresos Studio GUI to modify ECU configuration parameters values.
  4. Generate the configuration files.

## Chapter 4

### Function calls to module

- [Function Calls during Start-up](#)
- [Function Calls during Shutdown](#)
- [Function Calls during Wake-up](#)

#### 4.1 Function Calls during Start-up

LIN shall be initialized during STARTUP phase of EcuM initialization. The API to be called for this is `Lin_43_LPUART_FLEXIO_Init()`. The MCU module should be initialized before LIN module initialization.

The Lin driver does not need OS Support except for ISR's. Hence, can be initialized either in STARTUP1 or STARTUP2 phase of EcuM initialization. This depends on the implementation, desired duration for STARTUP1 & Target hardware design.

The LIN module shall be initialized by `Lin_43_LPUART_FLEXIO_Init(<&Lin_Configuration>)` service call during the start-up before the LIN peripherals are used.

Please note that GPIO pins used for connection of LIN physical layer have to be properly assigned to desired LPUART module prior the LIN initialization: so the MCU and PORT modules shall be initialized before LIN is initialized. After the LIN module is initialized each LIN channel has to be initialized as well before using it. This is also done by the `Lin_43_LPUART_FLEXIO_Init(<&Lin_Configuration>)` service.

In case FLEXIO channel is used, `Mcl_Init(<&Mcl_configuration>)` must be called before the lin driver init.

#### 4.2 Function Calls during Shutdown

There is no shutdown specific procedure for Lin driver. LIN driver can go to sleep mode using the following API services:

- `Lin_43_LPUART_FLEXIO_GoToSleepInternal(LIN_CHANNEL)`: which put the LIN driver into sleep mode without sending of Go-to-sleep command over the bus.
- `Lin_43_LPUART_FLEXIO_GoToSleep(LIN_CHANNEL)`: which put the LIN driver into sleep mode and send

#### 4.3 Function Calls during Wake-up

LIN driver supports the transmission of wake up command via the LIN bus.

For this purposes the `Lin_43_LPUART_FLEXIO_Wakeup(LIN_CHANNEL)` API service may be used.

#### 4.3.1 External Wakeup:

The Lin driver supports external wake-up from the bus in 2 modes:

Upon wakeup detection on Rx pin of the configured LPUART channel the ISR "LIN\_LPUARTx\_RxTx\_IRQHandler()" will be executed (based on the LIN Channel configured) to wake-up Lin Driver.

Upon wakeup detection on Rx pin of the configured FLEXIO channel the ISR "Flexio\_Lin\_Ip\_IRQHandler()" will be executed (based on the LIN Channel configured) to wake-up Lin Driver.

## Chapter 5

### Module requirements

- Exclusive areas to be defined in BSW scheduler
- Exclusive areas not available on this platform
- Peripheral Hardware Requirements
- ISR to configure within AutosarOS - dependencies
- ISR Macro
- Other AUTOSAR modules - dependencies
- Data Cache Restrictions
- User Mode support
- MultiPartition support

#### 5.1 Exclusive areas to be defined in BSW scheduler

In the current implementation, LIN is using the services of Schedule Manager (SchM) for entering and exiting the critical regions, to preserve a resource. SchM implementation is done by the integrators of the MCAL using OS or non-OS services. For testing the LIN, stubs are used for SchM. The following critical regions are used in the LIN driver: **Exclusive Areas are used in High level driver layer (HLD)**

- **LIN\_EXCLUSIVE\_AREA\_08** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the CTRL register from the write operation in `Lpuart_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the for BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_AbortTransferData`.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_GoToSleep` to protect the for BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_AbortTransferData`.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_AbortTransferData`.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the for BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_GoToSleepMode`.

- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the BAUD register from the write operation in `Lpuart_Lin_Ip_SendFrame`.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_WakeUpInternal` to protect the for BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the for BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_10** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the STAT register from the write operation in `Lpuart_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_10** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_AbortTransferData`.
- **LIN\_EXCLUSIVE\_AREA\_10** is used in function `Lin_43_LPUART_FLEXIO_GoToSleep` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_AbortTransferData`.
- **LIN\_EXCLUSIVE\_AREA\_10** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_AbortTransferData`.
- **LIN\_EXCLUSIVE\_AREA\_11** is used in function `Lin_43_LPUART_FLEXIO_GoToSleep` to protect the for TIMCMP register from the write operation in `Flexio_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_11** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for TIMCMP register from the write operation in `Flexio_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_11** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the for TIMCMP register from the write operation in `Flexio_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_13** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the SHIFCTL, TIMCTL, SHIFTCFG, SHIFCTL, TIMCFG registers from the write operation in `Flexio_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_15** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the for TIMCTL register from the write operation in `Flexio_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_15** is used in function `Lin_43_LPUART_FLEXIO_GoToSleep` to protect the for TIMCTL register from the write operation in `Flexio_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_15** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for TIMCTL register from the write operation in `Flexio_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_16** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the TIMCMP, TIMCTL registers from the write operation in `Flexio_Lin_Ip_SendFrame`.
- **LIN\_EXCLUSIVE\_AREA\_17** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the for TIMCTL register from the write operation in `Flexio_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_17** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for TIMCTL register from the write operation in `Flexio_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_17** is used in function `Flexio_Lin_Ipw_Callback` to protect the for TIMCTL register from the write operation in `Flexio_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_18** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the for TIMCTL, TIMCMP registers from the write operation in `Flexio_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_18** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for TIMCTL, TIMCMP registers from the write operation in `Flexio_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_18** is used in function `Flexio_Lin_Ipw_Callback` to protect the for TIMCTL, TIMCMP registers from the write operation in `Flexio_Lin_Ip_GoToSleepMode`.



- **LIN\_EXCLUSIVE\_AREA\_20** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the CTRL register from the write operation in `Lpuart_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_21** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the CTRL register from the write operation in `Lpuart_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_22** is used in function `Lin_43_LPUART_FLEXIO_SetClockMode` to protect the updates for BAUD register.
- **LIN\_EXCLUSIVE\_AREA\_23** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the BAUD register from the write operation in `Lpuart_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_24** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_24** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_24** is used in function `Lin_43_LPUART_FLEXIO_Ipw_Callback` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_24** is used in function `Lin_43_LPUART_FLEXIO_WakeUpInternal` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_24** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the STAT register from the write operation in `Lpuart_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the STAT register from the write operation in `Lpuart_Lin_Ip_SendFrame`.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lin_43_LPUART_FLEXIO_GoToSleepInternal` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lin_43_LPUART_FLEXIO_Ipw_Callback` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GoToSleepMode`.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lin_43_LPUART_FLEXIO_SendFrame` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lin_43_LPUART_FLEXIO_WakeUpInternal` to protect the for STAT register from the write operation in `Lpuart_Lin_Ip_GotoIdleState`.
- **LIN\_EXCLUSIVE\_AREA\_26** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the BAUD register from the write operation in `Lpuart_Lin_Ip_Init`.
- **LIN\_EXCLUSIVE\_AREA\_27** is used in function `Lin_43_LPUART_FLEXIO_Init` to protect the BAUD, CTRL registers from the write operation in `Lpuart_Lin_Ip_Init`.

#### Exclusive Areas implemented in Low level driver layer (IPL)

- **LIN\_EXCLUSIVE\_AREA\_08** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for CTRL register.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lpuart_Lin_Ip_SendFrame` to protect the updates for BAUD register.

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- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for BAUD, CTRL registers.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lpuart_Lin_Ip_Deinit` to protect the updates for BAUD, CTRL registers.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lpuart_Lin_Ip_AbortTransferData` to protect the updates for BAUD, CTRL registers.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lpuart_Lin_Ip_GoToSleepMode` to protect the updates for BAUD, CTRL registers.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lpuart_Lin_Ip_GotoIdleState` to protect the updates for BAUD, CTRL registers.
- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `Lpuart_Lin_Ip_AutoBaudCapture` to protect the updates for CTRL register.
- **LIN\_EXCLUSIVE\_AREA\_10** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_10** is used in function `Lpuart_Lin_Ip_AbortTransferData` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_11** is used in function `Flexio_Lin_Ip_GotoIdleState` to protect the updates for TIMCMP register.
- **LIN\_EXCLUSIVE\_AREA\_13** is used in function `Flexio_Lin_Ip_Init` to protect the updates for SHIFCTL, TIMCTL, SHIFTCFG, SHIFTCFL, TIMCFG registers.
- **LIN\_EXCLUSIVE\_AREA\_15** is used in function `Flexio_Lin_Ip_GotoIdleState` to protect the updates for TIMCTL register.
- **LIN\_EXCLUSIVE\_AREA\_16** is used in function `Flexio_Lin_Ip_SendFrame` to protect the updates for TIMCMP, TIMCTL registers.
- **LIN\_EXCLUSIVE\_AREA\_17** is used in function `Flexio_Lin_Ip_GoToSleepMode` to protect the updates for TIMCTL register.
- **LIN\_EXCLUSIVE\_AREA\_18** is used in function `Flexio_Lin_Ip_GoToSleepMode` to protect the updates for TIMCTL, TIMCMP registers.
- **LIN\_EXCLUSIVE\_AREA\_19** is used in function `Flexio_Lin_Ip_Deinit` to protect the updates for SHIFCTL, TIMCTL registers.
- **LIN\_EXCLUSIVE\_AREA\_20** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for CTRL register.
- **LIN\_EXCLUSIVE\_AREA\_20** is used in function `Lpuart_Lin_Ip_Deinit` to protect the updates for CTRL register.
- **LIN\_EXCLUSIVE\_AREA\_21** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for CTRL register.
- **LIN\_EXCLUSIVE\_AREA\_21** is used in function `Lpuart_Lin_Ip_Deinit` to protect the updates for CTRL register.
- **LIN\_EXCLUSIVE\_AREA\_22** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for BAUD register.
- **LIN\_EXCLUSIVE\_AREA\_22** is used in function `Lpuart_Lin_Ip_AutoBaudCapture` to protect the updates for BAUD register.

- **LIN\_EXCLUSIVE\_AREA\_23** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for BAUD register.
- **LIN\_EXCLUSIVE\_AREA\_24** is used in function `Lpuart_Lin_Ip_GoToSleepMode` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_24** is used in function `Lpuart_Lin_Ip_GotoIdleState` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lpuart_Lin_Ip_SendFrame` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lpuart_Lin_Ip_AutoBaudCapture` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lpuart_Lin_Ip_GoToSleepMode` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `Lpuart_Lin_Ip_GotoIdleState` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_26** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for BAUD register.
- **LIN\_EXCLUSIVE\_AREA\_27** is used in function `Lpuart_Lin_Ip_Init` to protect the updates for BAUD, CTRL registers.

#### Exclusive Areas are used in Interrupt service request (ISR)

- **LIN\_EXCLUSIVE\_AREA\_09** is used in function `ISR(Lpuart_Lin_Ip_IRQHandler)` to protect the updates for BAUD, CTRL registers.
- **LIN\_EXCLUSIVE\_AREA\_10** is used in function `ISR(Lpuart_Lin_Ip_IRQHandler)` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_11** is used in function `ISR(Flexio_Lin_Ip_IrqHandler)` to protect the updates for TIMCMP register.
- **LIN\_EXCLUSIVE\_AREA\_15** is used in function `ISR(Flexio_Lin_Ip_IrqHandler)` to protect the updates for TIMCTL register.
- **LIN\_EXCLUSIVE\_AREA\_24** is used in function `ISR(Lpuart_Lin_Ip_IRQHandler)` to protect the updates for STAT register.
- **LIN\_EXCLUSIVE\_AREA\_25** is used in function `ISR(Lpuart_Lin_Ip_IRQHandler)` to protect the updates for STAT register.

| Exclusive Area Matrix |  | LIN_EXCLUSIVE_AREA_08 | LIN_EXCLUSIVE_AREA_09 | LIN_EXCLUSIVE_AREA_10 | LIN_EXCLUSIVE_AREA_11 | LIN_EXCLUSIVE_AREA_12 | LIN_EXCLUSIVE_AREA_13 | LIN_EXCLUSIVE_AREA_14 | LIN_EXCLUSIVE_AREA_15 | LIN_EXCLUSIVE_AREA_16 | LIN_EXCLUSIVE_AREA_17 | LIN_EXCLUSIVE_AREA_18 | LIN_EXCLUSIVE_AREA_19 | LIN_EXCLUSIVE_AREA_20 | LIN_EXCLUSIVE_AREA_21 | LIN_EXCLUSIVE_AREA_22 | LIN_EXCLUSIVE_AREA_23 | LIN_EXCLUSIVE_AREA_24 | LIN_EXCLUSIVE_AREA_25 | LIN_EXCLUSIVE_AREA_26 | LIN_EXCLUSIVE_AREA_27 |
|-----------------------|--|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exclusive Area ID     |  |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_08 |  | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     | x                     |                       |                       |                       |                       |                       | x                     |
| LIN_EXCLUSIVE_AREA_09 |  | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     | x                     | x                     | x                     |                       |                       | x                     | x                     |
| LIN_EXCLUSIVE_AREA_10 |  |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     | x                     |                       |                       |
| LIN_EXCLUSIVE_AREA_11 |  |                       |                       |                       |                       |                       |                       | x                     |                       | x                     |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_12 |  |                       |                       |                       |                       |                       |                       | x                     |                       | x                     |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_13 |  |                       |                       |                       |                       |                       |                       |                       | x                     | x                     | x                     | x                     | x                     |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_14 |  |                       |                       | x                     | x                     |                       |                       |                       |                       | x                     |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_15 |  |                       |                       |                       |                       | x                     |                       |                       |                       | x                     | x                     | x                     | x                     |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_16 |  |                       |                       | x                     | x                     | x                     | x                     |                       | x                     |                       | x                     | x                     | x                     |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_17 |  |                       |                       |                       |                       | x                     |                       |                       | x                     | x                     |                       | x                     | x                     |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_18 |  |                       |                       | x                     | x                     | x                     | x                     |                       | x                     | x                     | x                     |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_19 |  |                       |                       |                       |                       | x                     |                       |                       | x                     | x                     | x                     | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_20 |  | x                     | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     |                       |                       |                       |                       |                       | x                     |
| LIN_EXCLUSIVE_AREA_21 |  | x                     | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     |                       |                       |                       |                       |                       |                       | x                     |
| LIN_EXCLUSIVE_AREA_22 |  |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     |                       |                       | x                     | x                     |
| LIN_EXCLUSIVE_AREA_23 |  |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     |                       |                       |                       | x                     | x                     |
| LIN_EXCLUSIVE_AREA_24 |  |                       |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     |                       |                       |
| LIN_EXCLUSIVE_AREA_25 |  |                       |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     |                       |                       |                       |
| LIN_EXCLUSIVE_AREA_26 |  |                       | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     | x                     |                       |                       |                       | x                     |
| LIN_EXCLUSIVE_AREA_27 |  | x                     | x                     |                       |                       |                       |                       |                       |                       |                       |                       |                       |                       | x                     | x                     | x                     | x                     |                       |                       | x                     |                       |

(Extracted table from RTD\_LIN\_EXCLUSIVE\_AREAS.xlsx)

## 5.2 Exclusive areas not available on this platform

List of exclusive areas which are not available on this platform (or blank if they're all available).

- LIN\_EXCLUSIVE\_AREA\_00
- LIN\_EXCLUSIVE\_AREA\_01
- LIN\_EXCLUSIVE\_AREA\_02
- LIN\_EXCLUSIVE\_AREA\_03
- LIN\_EXCLUSIVE\_AREA\_04
- LIN\_EXCLUSIVE\_AREA\_05
- LIN\_EXCLUSIVE\_AREA\_06
- LIN\_EXCLUSIVE\_AREA\_07

## 5.3 Peripheral Hardware Requirements

The LIN physical interface should be connected to the LIN module pins in order to get the LIN bus voltage levels.

## 5.4 ISR to configure within AutosarOS - dependencies

The following ISR's are used by the LIN driver:

*S32K11X derivatives:*

| ISR Name                        | IRQ Number | Observations |
|---------------------------------|------------|--------------|
| LPUART_LIN_IP_RxTx_0_IRQHandler | 31         | None         |
| LPUART_LIN_IP_RxTx_1_IRQHandler | 30         | None         |
| MCL_FLEXIO_ISR                  | 25         | None         |

*S32K14X derivatives(except S32K142):*

| ISR Name                        | IRQ Number | Observations |
|---------------------------------|------------|--------------|
| LPUART_LIN_IP_RxTx_0_IRQHandler | 31         | None         |
| LPUART_LIN_IP_RxTx_1_IRQHandler | 33         | None         |
| LPUART_LIN_IP_RxTx_2_IRQHandler | 35         | None         |
| MCL_FLEXIO_ISR                  | 69         | None         |

*S32K142 derivative:*

| ISR Name                        | IRQ Number | Observations |
|---------------------------------|------------|--------------|
| LPUART_LIN_IP_RxTx_0_IRQHandler | 31         | None         |
| LPUART_LIN_IP_RxTx_1_IRQHandler | 33         | None         |
| MCL_FLEXIO_ISR                  | 69         | None         |

*S32M24X derivatives:*

| ISR Name                        | IRQ Number | Observations |
|---------------------------------|------------|--------------|
| LPUART_LIN_IP_RxTx_0_IRQHandler | 31         | None         |
| LPUART_LIN_IP_RxTx_1_IRQHandler | 33         | None         |
| MCL_FLEXIO_ISR                  | 69         | None         |

## 5.5 ISR Macro

RTD drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions.

### 5.5.1 Without an Operating System

The macro `USING_OS_AUTOSAROS` must not be defined.

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### 5.5.1.1 Using Software Vector Mode

The macro `USE_SW_VECTOR_MODE` must be defined and the ISR macro is defined as: `#define ISR(IsrName) void IsrName(void)`

In this case, the drivers' interrupt handlers are normal C functions and their prologue/epilogue will handle the context save and restore.

### 5.5.1.2 Using Hardware Vector Mode

The macro `USE_SW_VECTOR_MODE` must not be defined and the ISR macro is defined as: `#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)`

In this case, the drivers' interrupt handlers must also handle the context save and restore.

## 5.5.2 With an Operating System

Please refer to your OS documentation for description of the ISR macro.

## 5.6 Other AUTOSAR modules - dependencies

**Base:** The BASE module contains the common files/definitions needed by all RTD modules.

**Mcu:** The MCU driver provides services for basic microcontroller initialization, power down functionality, reset and microcontroller specific functions required by other RTD software modules. The clocks need to be initialized prior to using the LIN driver. This module is required for setting the "LIN Clock Reference" value.

**Port:** The PORT module is used to configure the port pins with the needed modes, before they are used by the LIN module. For each LINFlex, the TXD and RXD signals need to be configured.

**Det:** The DET module is used for enabling Development error detection. The API function used is `Det_ReportError()`. The activation/deactivation of Development error detection is configurable using the 'LinDevErrorDetect' configuration parameter.\*

**Dem:** The DEM module is used for enabling reporting of production relevant error status. The API function used is `Dem_ReportErrorStatus()`.

**EcuM:** This module is used for processing the Wakeup notifications of LIN. Whenever the module is in 'Sleep' mode and a wakeup event occurs on a wakeup capable channel, it is reported to EcuM through the `EcuM_CheckWakeupEvent()` API. This is configurable using the 'LinChannelWakeupInfo' configuration parameter.

**Resource:** RESOURCE module is used to select microcontroller's derivatives.

**Rte:** The RTE module is needed for implementing data consistency of exclusive areas that are used by LIN module.

**Platform:** The PLATFORM module is needed for managing interrupt, system.

**EcuC:** The ECUC module is used to manage partitions in multi-partition.

**OS:** The OS module is used to map each partition to core.

**LinIf:** The LINIF module is used to manage Lin bus.

**Mcl:** The Mcl module is used to enable the FLEXIO module, to access the common registers and to route the interrupts to the Lin driver.

## 5.7 Data Cache Restrictions

None

## 5.8 User Mode support

- [User Mode configuration in the module](#)
- [User Mode configuration in AutosarOS](#)

### 5.8.1 User Mode configuration in the module

No special measures need to be taken to run LIN module from user mode. The LIN driver code can be executed at any time from both supervisor and user mode.

### 5.8.2 User Mode configuration in AutosarOS

When User mode is enabled, the driver may have the functions that need to be called as trusted functions in AutosarOS context. Those functions are already defined in driver and declared in the header `<IpName>_IpTrustedFunctions.h`. This header also included all headers files that contains all types definition used by parameters or return types of those functions. Refer the chapter [User Mode configuration in the module](#) for more detail about those functions and the name of header files they are declared inside. Those functions will be called indirectly with the naming convention below in order to AutosarOS can call them as trusted functions.

```
Call_<Function_Name>_TRUSTED(parameter1,parameter2,...)
```

That is the result of macro expansion `OsIf_Trusted_Call` in driver code:

```
#define OsIf_Trusted_Call[1-6params](name,param1,...,param6) Call_##name##_TRUSTED(param1,...,param6)
```

So, the following steps need to be done in AutosarOS:

- Ensure `MCAL_ENABLE_USER_MODE_SUPPORT` macro is defined in the build system or somewhere global.
- Define and declare all functions that need to call as trusted functions follow the naming convention above in Integration/User code. They need to visible in `Os.h` for the driver to call them. They will do the marshalling of the parameters and call `CallTrustedFunction()` in OS specific manner.
- `CallTrustedFunction()` will switch to privileged mode and call `TRUSTED_<Function_Name>()`.
- `TRUSTED_<Function_Name>()` function is also defined and declared in Integration/User code. It will un-marshalling of the parameters to call `<Function_Name>()` of driver. The `<Function_Name>()` functions are already defined in driver and declared in `<IpName>_IpTrustedFunctions.h`. This header should be included in OS for OS call and indexing these functions.

## 5.9 MultiPartition support

1. The LIN implements the "MCAL Distribution" according to type II, in which the mappable element is set to Channel. For additional details, please refer to `AUTOSAR_EXP_BSWDistributionGuide`.
2. The Lin and the mappable elements can be allocated to zero, one or several ECUC partitions, by means of "LinEcucPartionRef". If the Lin is mapped to zero ECUC partitions, the Lin behavior reverts to single-partition implementation, similar to previous Autosar versions. If the Lin is mapped to one or more ECUC partitions, the Lin enforces the following multi-partition assumptions:
  - The LIN assumes there is a single/multiple EcucPartition allocated per core. Internally, the module will use the Partition ID returned by `GetPartitionID` API to reference the appropriate global data and configuration elements.
  - The LIN assumes the EcucPartitionIDs are defined in a compact/consecutive order, starting from zero. The rationale is that the number of EcucPartitions is used for dimensioning the LIN internal variables

## Module requirements

and the EcucPartitionIDs are used for indexing those variables. (AR-86601 Zero based and dense IDs for OS-Cores and OS-Applications).

- The LIN assumes that initialization is performed on each partition, `Lin_Init()` is called separately for each partition, using a different configuration structure.
- The LIN initialization expects the upper layer will pass the correct initialization pointer, specific to the partition in which the driver is to be used. For example: `EcucPartition_1` is assigned to PartitionID 1; `Lin_Init` function will be called with `Lin_Config_EcucPartition_1` configuration structure, on partition 1.
- The LIN will check upon each API call if the requested resource is configured to be available on the current partition, if DET error reporting is enabled.
- The LIN requires that all variables in NonCacheable MemMap sections be allocated accordingly, to avoid data corruption in multipartition context.
- The LIN assumes that RTE module implements the EXCLUSIVE AREAS to be partition-aware only. The rationale is that the module implementation ensures data integrity by separating the mappable elements for different partitions already, thus implementing the EXCLUSIVE AREAS in a blocking manner (ex: spin-lock) on a multipartition scope, might affect the performance of the drivers on the two partitions, although they might access separate HW elements. For single-partition scope, the EXCLUSIVE AREAS keep the same purpose as on previous AUTOSAR implementations. (\* - to be updated per Lin usecase, to be detailed/removed if some modules require such kind of functionality for critical features which cannot be atomically shared among partitions).
- The LIN assumes that each interrupt is routed by the system only to the core on which is supposed to be serviced.



## Chapter 6

### Main API Requirements

- [Main function calls within BSW scheduler](#)
- [API Requirements](#)
- [Calls to Notification Functions, Callbacks, Callouts](#)

#### 6.1 Main function calls within BSW scheduler

None.

#### 6.2 API Requirements

None.

#### 6.3 Calls to Notification Functions, Callbacks, Callouts

##### 6.3.1 Call-back Notifications

The LIN Driver uses some callback functions which have to be provided by the respective module:

- EcuM\_SetWakeupEvent and EcuM\_CheckWakeup have to be provided by the EcuM module.
- LinIf\_WakeupConfirmation, LinIf\_TxConfirmation, LinIf\_RxIndication, LinIf\_LinErrorIndication and LinIf\_HeaderIndication have to be provided by the LinIf module.

##### 6.3.2 User Notification

None

## Chapter 7

### Memory allocation

- Sections to be defined in [Lin\\_43\\_LPUART\\_FLEXIO\\_MemMap.h](#)
- Linker command file

#### 7.1 Sections to be defined in [Lin\\_43\\_LPUART\\_FLEXIO\\_MemMap.h](#)

| Section name                                                | Type of section    | Description                                                                                                                                                                                                                                           |
|-------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_CONFIG_DATA_UNSPECIFIED | Configuration Data | Start of Memory Section for Data Configuration                                                                                                                                                                                                        |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_CONFIG_DATA_UNSPECIFIED  | Configuration Data | End of Memory Section for Data Configuration                                                                                                                                                                                                          |
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_CODE                    | Code               | Start of memory Section for Code                                                                                                                                                                                                                      |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_CODE                     | Code               | Stop of memory Section for Code                                                                                                                                                                                                                       |
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_VAR_CLEARED_8           | Variables          | Used for variables which have to be aligned to 8 bits. For instance used for variables of size 8 bits or used for composite data types: arrays, structs containing elements of maximum 8 bits. These variables are cleared to zero by start-up code.  |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_VAR_CLEARED_8            | Variables          | End of the above section.                                                                                                                                                                                                                             |
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_VAR_CLEARED_32          | Variables          | Used for variables which have to be aligned to 8 bits. For instance used for variables of size 8 bits or used for composite data types: arrays, structs containing elements of maximum 32 bits. These variables are cleared to zero by start-up code. |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_VAR_CLEARED_32           | Variables          | End of the above section.                                                                                                                                                                                                                             |

| Section name                                                | Type of section | Description                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_VAR_CLEARED_BOOLEAN     | Variables       | Used for variables which have to be aligned to 8 bits. For instance used for variables of size 8 bits or used for composite data types: arrays, structs containing elements of boolean. These variables are cleared to zero by start-up code.                |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_VAR_CLEARED_BOOLEAN      | Variables       | End of the above section.                                                                                                                                                                                                                                    |
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_VAR_CLEARED_UNSPECIFIED | Variables       | Used for variables, structures, arrays when the SIZE (alignment) does not fit the criteria of boolean, 8, 16 or 32 bits. These variables are cleared to zero by start-up code.                                                                               |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_VAR_CLEARED_UNSPECIFIED  | Variables       | End of the above section.                                                                                                                                                                                                                                    |
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_VAR_INIT_8              | Variables       | Used for variables which have to be aligned to 8 bits. For instance used for variables of size 8 bits or used for composite data types: arrays, structs containing elements of maximum 8 bits. These variables are initialized with values after every reset |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_VAR_INIT_8               | Variables       | End of the above section.                                                                                                                                                                                                                                    |
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_VAR_INIT_UNSPECIFIED    | Variables       | Used for variables, structures, arrays when the SIZE (alignment) does not fit the criteria of boolean, 8, 16 or 32 bits. These variables are initialized with values after every reset.                                                                      |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_VAR_INIT_UNSPECIFIED     | Variables       | End of the above section.                                                                                                                                                                                                                                    |
| LIN_43_LPUART_FLEXIO_START_↔<br>SEC_CONST_UNSPECIFIED       | Constant Data   | Used for constants when the SIZE (alignment) does not fit the criteria of 8, 16 or 32 bits.                                                                                                                                                                  |
| LIN_43_LPUART_FLEXIO_STOP_S↔<br>EC_CONST_UNSPECIFIED        | Constant Data   | End of the above section.                                                                                                                                                                                                                                    |

## 7.2 Linker command file

Memory shall be allocated for every section defined in the driver's "<Module>"\_MemMap.h.

## Chapter 8

### Integration Steps

This section gives a brief overview of the steps needed for integrating this module:

1. Generate the required module configuration(s). For more details refer to section [Files Required for Compilation](#)
2. Allocate the proper memory sections in the driver's memory map header file ("`<Module>_MemMap.h`") and linker command file. For more details refer to section [Sections to be defined in `<Module>\_MemMap.h`](#)
3. Compile & build the module with all the dependent modules. For more details refer to section [Building the Driver](#)

## Chapter 9

### External assumptions for driver

The section presents requirements that must be complied with when integrating the LIN driver into the application.

| External Assumption Req ID | External Assumption Text                                                                                                                                                                                                                                                                              |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SWS_Lin_00045              | One LIN driver provides access to one LIN hardware unit type (simple U↔ART or dedicated LIN hardware) that may consist of several LIN channels.                                                                                                                                                       |
| SWS_Lin_00225              | There must be at least one statically defined configuration set available for the LIN driver. When the EcuM invokes the initialization function, it has to provide a specific pointer to the configuration that it wishes to use.                                                                     |
| SWS_Lin_00014              | Each LIN PID shall be associated with a checksum model (either 'enhanced' where the PID is included in the checksum, or 'classic' where only the response data is check-summed) (see Lin_PduType).                                                                                                    |
| SWS_Lin_00015              | Each LIN PID shall be associated with a response data length in bytes (see Lin_PduType).                                                                                                                                                                                                              |
| SWS_Lin_00210              | For LIN Master nodes, the upper layer of the LIN Driver has to keep the buffer data consistent until return of function call. For LIN Slave nodes, the upper layer of the LIN Driver has to keep the buffer data consistent until end of response transmission.                                       |
| SWS_Lin_00245              | The content of Lin_GeneralTypes.h shall be protected by a LIN_GENE↔RAL_TYPES define.                                                                                                                                                                                                                  |
| SWS_Lin_00246              | If different LIN drivers are used, only one instance of this file has to be included in the source tree. For implementation all Lin_GeneralTypes.h related types in the documents mentioned before shall be considered.                                                                               |
| SWS_Lin_00228              | Name: Lin_FramePidTypeKind: TypeDerived from: uint8Range: 0...0x↔FE: -: The LIN identifier (0...0x3F) together with its two parity bits.↔Description: Represents all valid protected identifier used by Lin_Send↔Frame().Available via: Lin_GeneralTypes.h Note: Under control of BASE module         |
| SWS_Lin_00229              | Name: Lin_FrameCsModelTypeKind: EnumerationRange: LIN_ENHA↔NCED_CS: -: Enhanced checksum modelLIN_CLASSIC_CS: -: Classic checksum modelDescription: This type is used to specify the Checksum model to be used for the LIN Frame.Available via: Lin_GeneralTypes.h Note: Under control of BASE module |

| External Assumption Req ID | External Assumption Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SWS_Lin_00230              | Name: Lin_FrameResponseTypeKind: EnumerationRange: LIN_FRAME←<br>ERESPONSE_TX: -: Response is generated from this nodeLIN_FRA←<br>MERESPONSE_RX: -: Response is generated from another node and is<br>relevant for this node.LIN_FRAMERESPONSE_IGNORE: -: Response is<br>generated from another node and is irrelevant for this nodeDescription←<br>: This type is used to specify whether the frame processor is required to<br>transmit the response part of the LIN frame.Available via: Lin_General←<br>Types.h Note: Under control of BASE module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| SWS_Lin_00231              | Name: Lin_FrameDlTypeKind: TypeDerived from: uint8Range: 1...8: -←<br>: Data length of a LIN FrameDescription: This type is used to specify<br>the number of SDU data bytes to copy.Available via: Lin_GeneralTypes.h<br>Note: Under control of BASE module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| SWS_Lin_00232              | Name: Lin_PduTypeKind: StructureElements: PidType: Lin_Frame←<br>PidTypeComment: -CsType: Lin_FrameCsModelTypeComment: -Drc←<br>Type: Lin_FrameResponseTypeComment: -DlType: Lin_FrameDlType←<br>Comment: -SduPtrType: uint8Comment: -Description: This Type is used<br>to provide PID, checksum model, data length and SDU pointer from the<br>LIN Interface to the LIN driver.Available via: Lin_GeneralTypes.h Note:<br>Under control of BASE module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| SWS_Lin_00233              | Name: Lin_StatusTypeKind: EnumerationRange: LIN_NOT_OK: -←<br>: LIN frame operation return value. Development or production error<br>occurredLIN_TX_OK: -: LIN frame operation return value. Successful<br>transmission.LIN_TX_BUSY: -: LIN frame operation return value. On-<br>going transmission (Header or Response).LIN_TX_HEADER_ERROR←<br>: -: LIN frame operation return value. Erroneous header transmission such<br>as:: - Mismatch between sent and read back data - Identifier parity error<br>or - Physical bus errorLIN_TX_ERROR: -: LIN frame operation return<br>value. Erroneous response transmission such as:: - Mismatch between sent<br>and read back data - Physical bus errorLIN_RX_OK: -: LIN frame oper-<br>ation return value. Reception of correct response.LIN_RX_BUSY: -: LIN<br>frame operation return value. Ongoing reception: at least one response byte<br>has been received, but the checksum byte has not been received.LIN_←<br>RX_ERROR: -: LIN frame operation return value. Erroneous response<br>reception such as:: - Framing error - Overrun error - Checksum error or -<br>Short responseLIN_RX_NO_RESPONSE: -: LIN frame operation return<br>value. No response byte has been received so far.LIN_OPERATIONAL←<br>: -: LIN channel state return value. Normal operation; the related LIN<br>channel is ready to transmit next header. No data from previous frame<br>available (e.g. after initialization)LIN_CH_SLEEP: -: LIN channel state<br>return value. Sleep state operation; in this state wake-up detection from<br>slave nodes is enabled.Description: LIN operation states for a LIN chan-<br>nel or frame, as returned by the API service Lin_GetStatus().Available via:<br>Lin_GeneralTypes.h |
| SWS_Lin_00106              | The Lin module's environment shall not call any function of the Lin module<br>before having called Lin_Init except Lin_GetVersionInfo.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| SWS_Lin_00193              | In case of receiving data the LIN Interface has to wait for the corresponding<br>response part of the LIN frame by polling with the function Lin_GetStatus()<br>after using the function Lin_SendFrame().                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| SWS_Lin_00194              | The Lin module's environment shall only call Lin_SendFrame on a channel<br>which is in state LIN_CH_OPERATIONAL or in one of the sub-states of<br>LIN_CH_OPERATIONAL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| External Assumption Req ID | External Assumption Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SWS_Lin_00239              | In case of errors during header transmission, it is up to the implementer how to handle these errors (stop/continue transmission) and to decide if the corresponding response is valid or not.                                                                                                                                                                                                                                                                                                              |
| SWS_Lin_00200              | The return states LIN_TX_OK, LIN_TX_BUSY, LIN_TX_HEADER↔_ERROR, LIN_TX_ERROR, LIN_RX_OK, LIN_RX_BUSY, LIN_R↔X_ERROR, LIN_RX_NO_RESPONSE and LIN_OPERATIONAL are sub-states of the channel state LIN_CH_OPERATIONAL.                                                                                                                                                                                                                                                                                         |
| SWS_Lin_CONSTR_00279       | LinChannel and LinTrcvChannel of one communication channel shall all reference the same ECUC partition.                                                                                                                                                                                                                                                                                                                                                                                                     |
| SWS_Lin_91140              | Name: Lin_SlaveErrorTypeKind: EnumerationRange: LIN_ERR_HEA↔DER: -: Error in headerLIN_ERR_RESP_STOPBIT: -: Framing error in responseLIN_ERR_RESP_CHKSUM: -: Checksum errorLIN_ERR_R↔ESP_DATABIT: -: Monitoring error of transmitted data bit in response↔LIN_ERR_NO_RESP: -: No responseLIN_ERR_INC_RESP: -: Incomplete responseDescription: This type represents the slave error types that are detected during header reception and response transmission / reception.↔Available via: Lin_GeneralTypes.h |
| EA_RTD_00042               | The application shall not preempt a LIN function working on a channel by calling the same or another LIN function targeting the same channel. Note: The following functions are targeted by the requirement:                                                                                                                                                                                                                                                                                                |

- Lin\_CheckWakeup()
- Lin\_GoToSleep()
- Lin\_GoToSleepInternal()
- Lin\_Wakeup()
- Lin\_GetStatus()
- Lin\_SendFrame()

EA\_RTD\_00043 | The application shall call the function Lin\_Init only once during runtime, as stated in SWS↔\_Lin\_00146. EA\_RTD\_00044 | Lin\_Init() function shall be called first to initialize the driver. Application shall not call any function of LIN API before the function Lin\_Init() has completed. Only Lin\_GetVersionInfo() can be called before Lin\_Init(). EA\_RTD\_00045 | The application shall only call Lin\_SendFrame on a channel which is in state LIN\_CH\_OPERATIONAL or in one of the sub-states of LIN\_CH\_OPERATIONAL. Note: **Rationale:** If Lin\_SendFrame() is called while the LIN channel state is LIN\_CH\_SLEEP, the module might lose its internal consistency. The same risk exists in case of calling Lin\_SendFrame() while module state is LIN\_NOT\_OK.

**Implementation Hint:** Before calling Lin\_SendFrame() the application shall call Lin\_GetStatus() in order to ensure that the module state is compliant.

EA\_RTD\_00071 | If interrupts are locked, a centralized function pair to lock and unlock interrupts shall be used. EA\_RTD\_00082 | When caches are enabled and data buffers are allocated in cacheable memory regions the buffers involved in DMA transfer shall be aligned with both start and end to cache line size. Note: **Rationale:** This ensures that no other buffers/variables compete for the same cache lines.

EA\_RTD\_00093 | The application shall define EcucCoreIDs in a compact/consecutive order, starting from zero. EA\_RTD\_00096 | The application shall pass the correct initialization pointer, specific to the partition in which the driver is to be used. EA\_RTD\_00106 | Standalone IP configuration and HL configuration of the same driver shall be done in the same project EA\_RTD\_00107 | The integrator shall use the IP interface only for hardware resources that were configured for standalone IP usage. Note: The integrator shall not directly use the IP interface for hardware resources that were allocated to be used in HL context. EA\_RTD\_00108 | The integrator shall use the IP interface to build a CDD, therefore the BSWMD will not contain reference to the IP interface EA\_RTD\_00113

## External assumptions for driver

| When RTD drivers are integrated with AutosarOS and User mode support is enabled, the integrator shall assure that the definition and declaration of all RTD functions needed to be called as trusted functions follow the naming convention `Call<Function_Name>TRUSTED(parameter1,parameter2,...)` in Integration/User code. They need to be visible in `Os.h` for the driver to call them. They will call `RTD <Function_Name>()` as trusted functions in OS specific manner.



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